MACHINE LEARNING

MODEL METRICS

Vehicle Classification (Sedan / SUV / Sports Car)

Github: https://github.com/ParthRawat1302/Machine-learning/tree/main/Vehicle-Classification

1. How to Read These Numbers

- **Accuracy**: share of vehicles whose class is predicted correctly, in %; higher is better
- **Clean**: the validation rows exactly as recorded. **Stress 0.15**: 0.15 sd of noise on every value, harsher than the test file
- **Test-like**: the validation rows carry the noise measured in the test file (0.08 sd on every value, or 0.12 sd on a random half; re-clipped and re-rounded), averaged over both readings. This is the selection metric
- **Gap vs final ± s.e.**: paired difference in test-like accuracy to the final model, in points, with one standard error; only gaps beyond about two s.e. are real
- **CV**: repeated stratified 5-fold (3 × 5 = 15 folds) on the 7,500 de-duplicated training rows, the same folds and noisy copies for every model; challengers are re-tested on 15 fresh folds
- Stacking and threshold tuning are cross-fitted (learned on four folds, applied to the fifth)

2. Summary

The final model is **logistic regression** (multinomial, C = 0.1, standardised features) trained on the training rows plus four noise-jittered copies. No other model, preprocessing or ensemble beats it.

Final model	Accuracy %
Clean validation rows (15 folds)	89.58
Test-like noise (15 folds), the selection metric	**89.26**
Test-like noise, 15 fresh confirmation folds	89.20
Stress: 0.15 sd noise on every value (15 folds)	88.96
Public leaderboard	about 90

Best configuration of each model family:

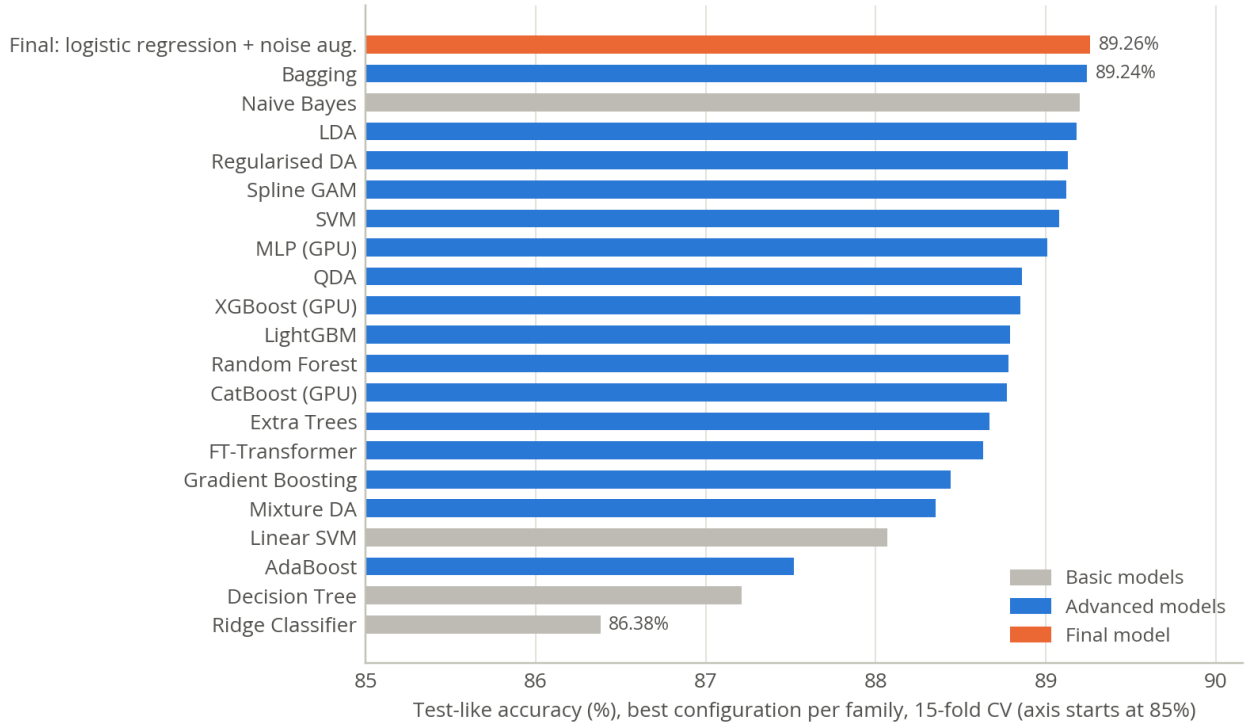


Figure 1. Best configuration of every model family, test-like accuracy, 15-fold CV (TabPFN ran on 5 folds only, see 4.1).

3. Basic Models (15-fold CV)

Model	Configuration	Clean	Test-like	Stress 0.15	Gap vs final	± s.e.
Logistic Regression	baseline: C = 1, no augmentation	89.44	89.20	88.92	−0.07	0.05
Logistic Regression	**C = 0.1 + noise augmentation (final)**	**89.58**	**89.26**	**88.96**	**0.00**	**0.00**
Logistic Regression	one-vs-rest, C = 0.1 + aug.	88.47	88.36	88.04	−0.90	0.09
Linear SVM	hinge loss, C = 0.05 + aug.	88.25	88.07	87.84	−1.19	0.10
Ridge Classifier	least squares, α = 10 + aug.	86.48	86.38	86.08	−2.88	0.16
Decision Tree	depth 6, min leaf 20	86.16	86.05	85.65	−3.21	0.12
Decision Tree	depth 10, min leaf 10	85.22	85.12	84.68	−4.14	0.15
Decision Tree	depth 4 + 2 LDA features	87.39	87.21	86.82	−2.05	0.14
Naive Bayes	Gaussian, raw features	84.67	84.64	84.53	−4.62	0.19
Naive Bayes	PCA whitening (10 comps) + aug.	86.26	86.12	85.86	−3.14	0.14
Naive Bayes	LDA(2) projection + aug.	89.36	89.20	88.84	−0.07	0.04

3.1 Tuning sweep

Inverse regularisation strength C, each with four noise-jittered training copies (columns). Without augmentation (C = 1) the test-like accuracy is 89.20%.

Logistic regression + augmentation	C = 0.01	C = 0.03	C = 0.1	C = 0.3	C = 1
Clean	89.45	89.52	89.58	89.55	89.54
Test-like	89.25	89.28	89.26	89.24	89.24
Stress 0.15	89.05	88.96	88.96	88.98	88.99
Gap vs final (± s.e.)	−0.02 (0.03)	+0.01 (0.02)	–	−0.02 (0.01)	−0.02 (0.01)

3.2 Final model: confusion matrix

Out-of-fold predictions of the first CV repeat under test-like noise (each of the 7,500 vehicles counted once):

True \ predicted	Sedan	SUV	Sports Car	Precision	Recall	F1
Sedan	2,150	218	170	0.845	0.847	0.846
SUV	222	2,258	10	0.908	0.907	0.907
Sports Car	172	11	2,288	0.927	0.926	0.927

4. Advanced Models (15-fold CV)

All 27 advanced configurations, sorted by test-like accuracy, with the final model in bold for reference. Test predictions come from a refit on all training rows. Logistic Regression with Yeo-Johnson transform + aug (+0.04 ± 0.02) screens above it by less than two standard errors; on the fresh folds a Yeo-Johnson model ties the final one (0.00, Section 7).

Model	Configuration	Clean	Test-like	Stress 0.15	Gap vs final	± s.e.
Logistic Regression	Yeo-Johnson transform + aug.	89.56	89.30	88.96	+0.04	0.02
Logistic Regression	**C = 0.1 + noise augmentation (final)**	**89.58**	**89.26**	**88.96**	**0.00**	**0.00**
Bagging	50 logistic regressions	89.46	89.24	88.93	−0.03	0.02
Logistic Regression	degree-2 polynomial, C = 0.01	89.42	89.20	88.95	−0.06	0.03
LDA	covariance shrinkage 0.1	89.35	89.18	88.87	−0.09	0.03
Regularised DA	α = 0.1, γ = 0.02	89.23	89.13	88.88	−0.13	0.03
Spline GAM	5 knots, C = 0.05 + aug.	89.37	89.12	88.76	−0.15	0.06
SVM	RBF kernel, C = 1	89.19	89.08	88.79	−0.18	0.05
MLP (GPU)	64 × 2, noise 0.2, dropout 0.1	89.22	89.01	88.80	−0.25	0.07
Spline GAM	5 knots, 5 test-noise copies	89.14	88.88	88.51	−0.39	0.06
QDA	reg_param 0.1	88.95	88.86	88.75	−0.41	0.06
XGBoost (GPU)	depth 2, 600 trees, LDA feats	89.00	88.85	88.56	−0.42	0.07
LightGBM	7 leaves, LDA feats, noise copies	88.84	88.79	88.50	−0.47	0.09
Random Forest	min leaf 5, LDA feats, noise copies	88.78	88.78	88.54	−0.49	0.06
CatBoost (GPU)	depth 4, 1,000 trees, LDA feats	88.79	88.77	88.54	−0.50	0.06
Extra Trees	500 trees, min leaf 5, LDA feats	88.78	88.67	88.42	−0.60	0.08
FT-Transformer	d = 32, 2 layers, noise 0.2	88.68	88.63	88.54	−0.63	0.09
Gradient Boosting	scikit-learn, 300 depth-2 trees	88.55	88.44	88.26	−0.83	0.10
MLP (GPU)	64 × 2, no regularisation	88.64	88.40	88.08	−0.86	0.09
Mixture DA	4 tied Gaussians per class	88.52	88.35	87.88	−0.92	0.08
Random Forest	500 trees	88.32	88.27	88.29	−1.00	0.08
XGBoost (GPU)	depth 2, 1,200 trees	88.38	88.22	87.97	−1.05	0.07
FT-Transformer	d = 64, 3 layers, noise 0.15	88.16	88.04	87.80	−1.22	0.11
MLP (GPU)	256 × 2, noise 0.2, dropout 0.1	88.15	87.99	87.64	−1.28	0.13
Gradient Boosting	histogram, depth 3, 500 rounds	87.97	87.92	87.64	−1.35	0.10
Bagging	300 decision trees	87.99	87.87	87.76	−1.40	0.08
MLP (GPU)	256 × 3, noise 0.3, dropout 0.1	87.99	87.87	87.61	−1.40	0.10

Model	Configuration	Clean	Test-like	Stress 0.15	Gap vs final	± s.e.
AdaBoost	300 depth-2 trees, rate 0.3	87.79	87.52	87.14	−1.74	0.09

4.1 TabPFN v2 (5 folds)

TabPFN is slow (about 80 s per fold on the RTX 4060 Laptop GPU), so it ran on the 5 folds of the first CV repeat; the final model on the same 5 folds for comparison:

Model	Configuration	Clean	Test-like	Stress 0.15	Gap vs final	± s.e.
Logistic Regression	C = 0.1 + noise augmentation (final)	89.57	89.36	88.99	0.00	0.00
TabPFN v2	pretrained transformer, 4 members	89.32	89.13	88.83	−0.23	0.08

5. Ensembles

Probability averages of the final model with the best model of other families, a stacked logistic meta-model and per-class threshold tuning (both cross-fitted).

Blend	Clean	Test-like	Stress 0.15	Gap vs final	± s.e.
Average: final + spline GAM (noise copies)	89.51	89.28	88.84	+0.01	0.05
Per-class threshold tuning (log-probability offsets)	89.42	89.25	88.92	−0.01	0.02
Average: final + polynomial logistic regression	89.48	89.24	88.95	−0.02	0.02
Average: final + LDA + Naive Bayes (LDA projection)	89.42	89.24	88.90	−0.02	0.03
Average: final + SVM (RBF)	89.39	89.24	88.91	−0.03	0.04
Average: final + MLP	89.44	89.22	89.05	−0.05	0.05
Average: final + random forest	89.23	89.19	88.84	−0.07	0.04
Average: final + XGBoost	89.31	89.17	88.87	−0.09	0.04
Average: final + CatBoost	89.28	89.11	88.82	−0.16	0.03
Stacking: logistic meta-model over 12 models	89.29	89.09	88.78	−0.17	0.06

6. Feature Engineering and Preprocessing (15-fold CV)

Each idea is placed in front of the final logistic regression and scored on the same folds. The best screening gap, Yeo-Johnson (+0.06 ± 0.02), shrinks to 0.00 on the fresh folds (Section 7): the winner's curse that the confirmation step exists to catch.

Idea (in front of the final model)	Clean	Test-like	Stress 0.15	Gap vs final	± s.e.
None: the final model (reference)	**89.58**	**89.26**	**88.96**	**0.00**	**0.00**
10 physics ratios (power/weight, hp per litre, ...)	89.45	89.24	88.96	−0.02	0.02
Degree-2 polynomial features (230)	89.42	89.20	88.95	−0.06	0.03
Spline expansion (GAM, 5 knots)	89.37	89.12	88.76	−0.15	0.06
Clip-limit indicator flags	89.46	89.21	88.86	−0.06	0.04
Yeo-Johnson power transform	89.61	89.32	88.95	+0.06	0.02
Rank-Gauss (quantile) transform	89.48	89.27	89.01	+0.01	0.04
Robust scaling (median / IQR)	89.57	89.27	88.94	+0.01	0.01
Un-censoring clipped values	89.59	89.28	88.97	+0.02	0.01
PCA denoising (17 components)	89.45	89.23	88.95	−0.04	0.03
Feature selection (top 17 by ANOVA)	88.79	88.75	88.44	−0.52	0.07
Lasso (L1) logistic regression	89.47	89.25	88.94	−0.01	0.03
Data cleaning (drop implausible rows)	89.45	89.20	88.96	−0.07	0.03
Training on 10 test-noise copies	89.48	89.21	89.00	−0.05	0.04

7. Fresh-Fold Confirmation (15 new folds)

The closest challengers re-run on folds nothing was tuned on (seeds 2024 / 99 / 314). A model would replace the final one only if it still won here by more than two standard errors.

Model	Clean	Test-like	Stress 0.15	Gap vs final	± s.e.
Logistic Regression, final	**89.34**	**89.20**	**88.78**	**0.00**	**0.00**
Naive Bayes, LDA(2) projection + aug.	89.30	89.18	89.01	−0.02	0.04
LDA, covariance shrinkage 0.1	89.33	89.18	88.89	−0.02	0.03
Logistic Regression, degree-2 polynomial, C = 0.01	89.35	89.19	88.75	−0.01	0.03
Spline GAM, 5 knots, 5 test-noise copies	89.07	88.90	88.51	−0.31	0.10
Yeo-Johnson power transform	89.32	89.20	88.81	0.00	0.02
Un-censoring clipped values	89.40	89.21	88.80	+0.01	0.02
Average: final + polynomial logistic regression	89.35	89.21	88.80	+0.01	0.02
Average: final + spline GAM (noise copies)	89.48	89.21	88.75	+0.01	0.05

8. Robustness to Measurement Noise (5 folds)

Accuracy (%) as noise of increasing size (columns, in feature standard deviations; re-clipped and re-rounded) is added to the validation rows. The test file carries about 0.08.

Model	0	0.05	0.1	0.15	0.2	0.3	0.4
Logistic regression	89.57	89.29	89.28	89.12	88.61	88.05	86.96
Naive Bayes + LDA(2)	89.35	89.23	89.15	89.04	88.79	87.89	87.11
QDA (reg 0.1)	88.95	88.96	88.76	88.68	88.43	87.48	86.91
Random forest	88.28	88.28	88.21	88.24	87.84	86.83	86.09
Decision tree (depth 6)	85.95	86.16	85.67	85.35	85.05	82.69	80.52
Gaussian Naive Bayes	84.65	84.69	84.84	84.69	84.44	83.85	84.05

9. Training-Set Size (5 folds)

Test-like accuracy (%) when each model is trained on a stratified 20% of every training fold (1,199 instead of 6,000 rows) and scored on the same noisy validation rows.

Model	100% of the data	20% of the data	Change
Logistic regression	89.36	88.78	−0.58
Naive Bayes + LDA(2)	89.22	89.00	−0.22
SVM (RBF, C=1)	89.10	88.52	−0.58
Neural network (MLP 64x2)	89.01	87.66	−1.35
XGBoost + LDA features	88.89	88.05	−0.84
QDA (reg 0.1)	88.77	88.51	−0.26
Random forest	88.26	87.72	−0.53
Decision tree (depth 6)	85.71	84.15	−1.56
Gaussian Naive Bayes	84.68	84.70	+0.02

10. Unsupervised Learning

Clustering on the standardised features without labels, compared with the true classes; *matched accuracy* is the accuracy after optimally matching clusters to classes.

Algorithm	Clusters	Noise points	Silhouette	ARI	Homogeneity	Matched accuracy %
K-Means (k=3)	3	0	0.293	0.532	0.484	81.35
Gaussian mixture (k=3)	3	0	0.141	0.356	0.369	71.29
Agglomerative (Ward, k=3)	3	0	0.228	0.378	0.354	71.83
DBSCAN (eps = 2.20)	1	58	–	0.000	0.000	33.63

11. Validation Checks

2,500 of the 10,000 training rows are exact duplicates. Left in, they inflate the accuracy of models that can memorise (%):

Model (plain 5-fold CV)	With duplicates	Without	Inflation
Gaussian Naive Bayes	84.39	84.65	−0.26
1-nearest-neighbour	89.06	82.97	+6.09
Decision tree (unlimited depth)	89.61	82.29	+7.32

The extra measurement error of the test file was measured, not guessed:

Train vs test check	Result
Test / train correlation ratio (58 pairs with $\|r\| > 0.6$)	0.9932
Noise that reproduces it in the training data	about 0.08 feature sd
Adversarial validation AUC, train vs test (0.5 = identical)	0.559 ± 0.006